

Application of Physics-Informed Neural Networks with Topological Validation on Gene Regulatory Networks

Wael El Khateeb, Bernardo Rivas, William Kalies
The University of Toledo

**Advanced Progresses in Population Models Driven by Natural Intelligence
and/or Artificial Intelligence-University of Central Florida (UCF)**



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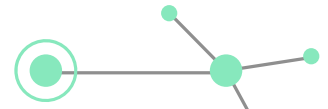
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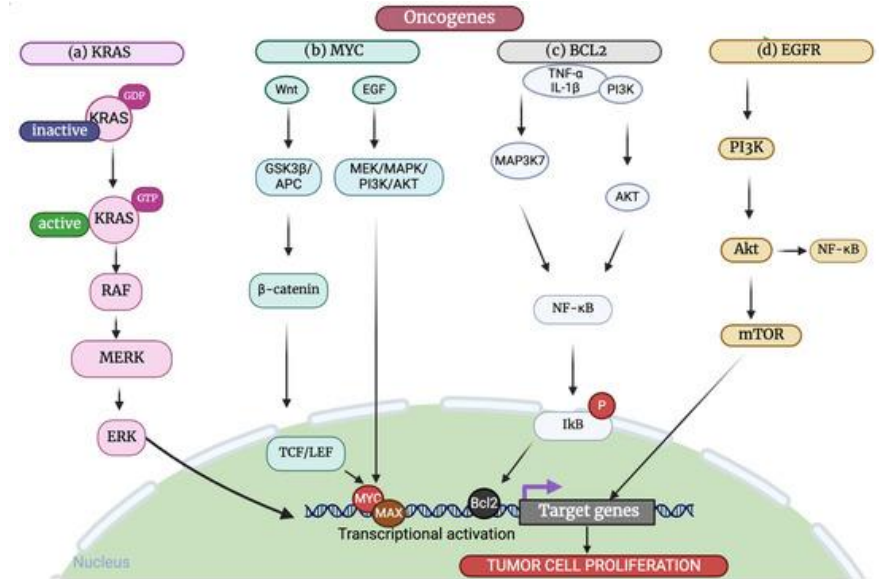
04 Results and insights to Epidemiology and Ecology



Cause of Cancer:

- 1) Tobacco
- 2) Infections
- 3) Radiation
- 4) Exposure to chemicals
- 5) Diets

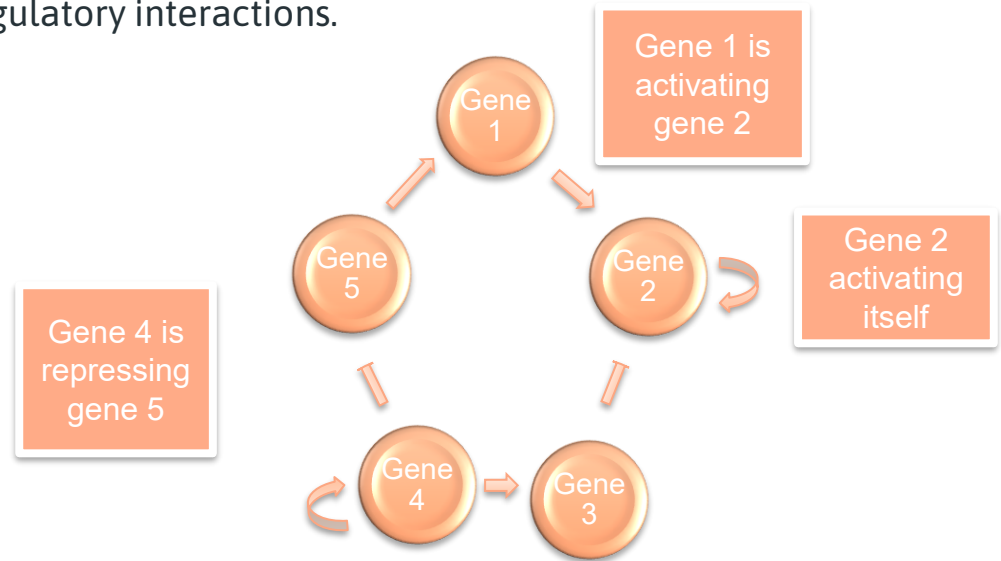
6) Genes!



Gene Regulatory Networks:

A **gene regulatory network (GRN)** is a representation of how genes control each other's activity through regulatory interactions.

Made of nodes and edges.



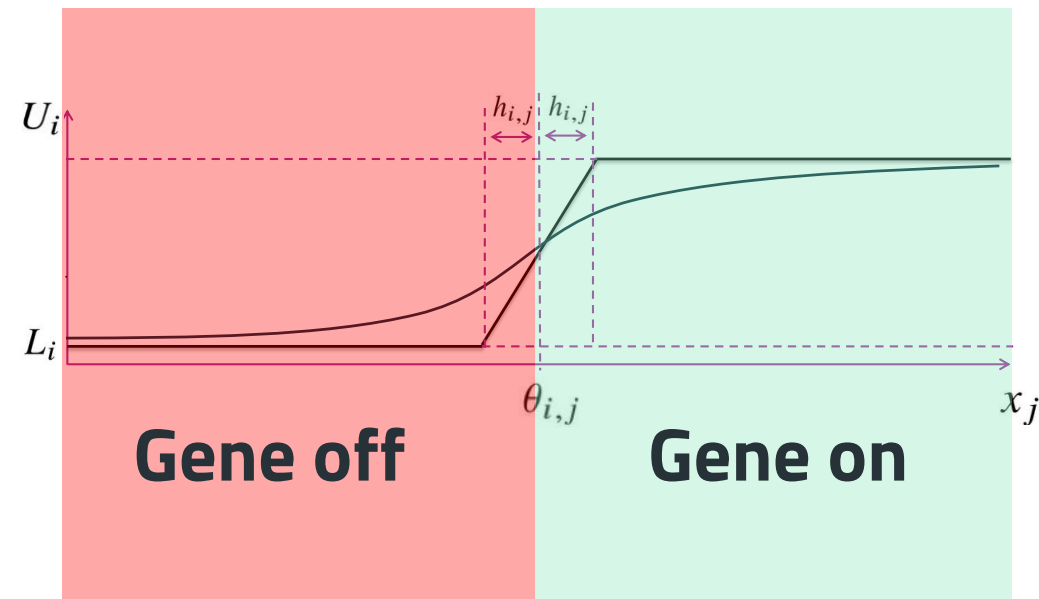
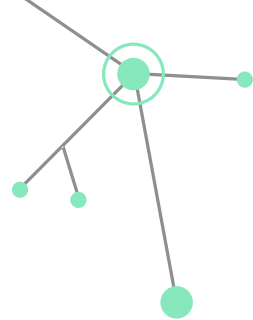
Formal definition

Gene regulatory network: Let N be a positive integer. We define a regulatory network of dimension N to be a graph consisting of vertices (or nodes) $V = [N]$ and signed directed edges $E \subseteq V \times V$. Each edge is assigned a sign, indicating whether the interaction between nodes is positive (activation) or negative (repression).

Define the target of a node i as $T(i) = \{j | (i, j) \in E\}$ and the Source of a node j as $S(j) = \{i | (i, j) \in E\}$

Gene-Gene Interactions

The effect of one gene on another is observed only when the relevant expression level crosses a threshold $\theta_{i,j}$. These interactions can be modeled using sigmoidal functions including:



$$g_i(x_j) = \begin{cases} U_i, & x_j > \theta_{i,j} + h_{i,j} \\ S_{i,j}(x_j), & \theta_{i,j} - h_{i,j} \leq x_j \leq \theta_{i,j} + h_{i,j} \\ L_i, & x < \theta_{i,j} - h_{i,j}, \end{cases}$$

$$f_i(x_j) = L_i + (U_i - L_i) \frac{\theta_j^d}{\theta_j^d + x_j^n}$$

How to model using ODES:

$$\frac{dx_j}{dt} = \boxed{Production_j(x_i(t))} - \boxed{Decay_j(x_j(t))}$$

$-d_j x_j$, provided that d_j is the **effective decay/dilution rate** for gene j 's product.

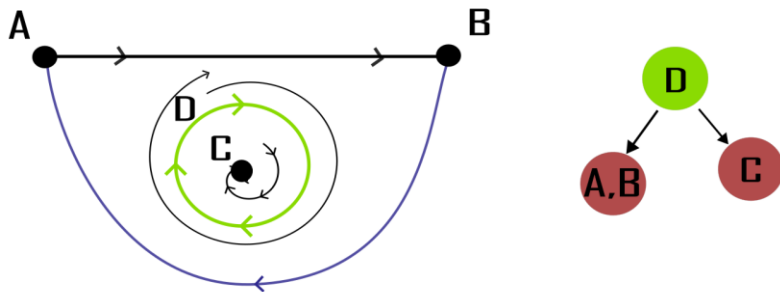
$$H_{ij}(x_i, U_{ij}, L_{ij}, \theta_{ij}) = L_{ij} + (U_{ij} - L_{ij})F_{ij}(x_i)$$

$$Production_j(x_i(t)) = \sum_{m=1}^{M_i} \prod_{i \in S_{jm}} H_{ij}(x_i)$$

Morse Decomposition:

Let X be a compact metric space and $\varphi: \mathbb{R}_{\geq 0} \times X \rightarrow X$ a continuous semiflow.

A family $\mathcal{M} = \{M_\alpha\}_{\alpha \in A}$ of nonempty, pairwise disjoint, compact, invariant sets is a **Morse decomposition** of X if there is a strict partial order $<$ on A such that for every $x \in X$ whose full orbit is defined forward and backward, either $x \in \bigcup_\alpha M_\alpha$ or there exist indices $\alpha \leq \beta$ with $\omega(x) \subset M_\alpha$, $\alpha \leq \beta$, $M_\beta \subset \alpha(x)$, where $\omega(x)$ and $\alpha(x)$ are the omega and alpha limit sets of x . All recurrent dynamics is contained in the M_α

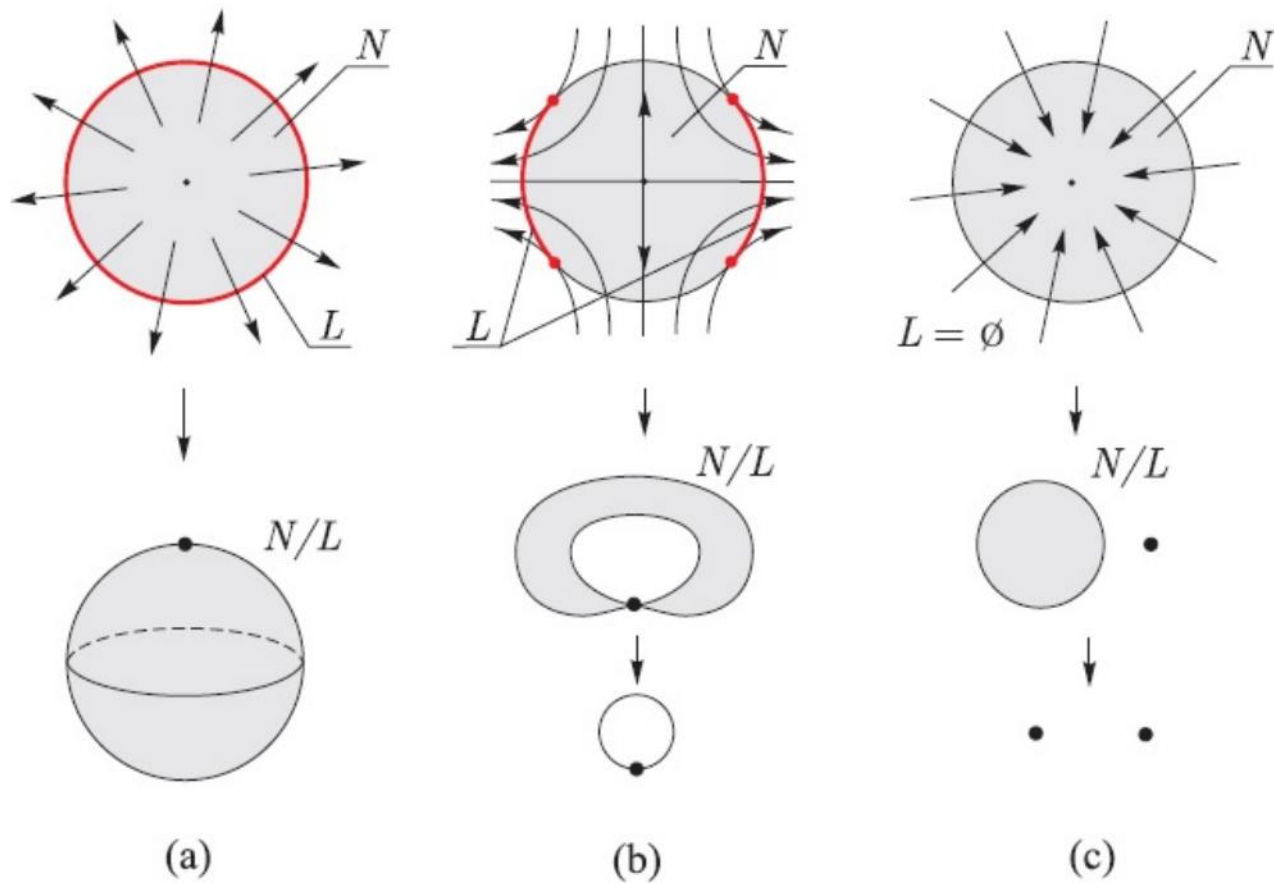


Let S be an isolated invariant set. A pair of compact sets (N, L) with $L \subset N$ is an index pair for S if:
Index pair conditions

1. Isolation of S : $S = \text{Inv}(\text{cl}(N \setminus L))$, $N \setminus L$ is a neighborhood of S .
2. Positive invariance of $L \subset N$: if $x \in L$ and $\varphi([0, t], x) \subset N$, then $\varphi([0, t], x) \subset L$.
3. Exit set property: if $x \in N$ and $t_0 > 0$ with $\varphi(t_0, x) \notin N$, then there exists $0 \leq t_1 < t_0$ such that $\varphi(t_1, x) \in L$.

The homotopy Conley index of S is the homotopy type of the pointed space obtained by collapsing L to a point:

$$h(S) \sim (N/L, [L]).$$



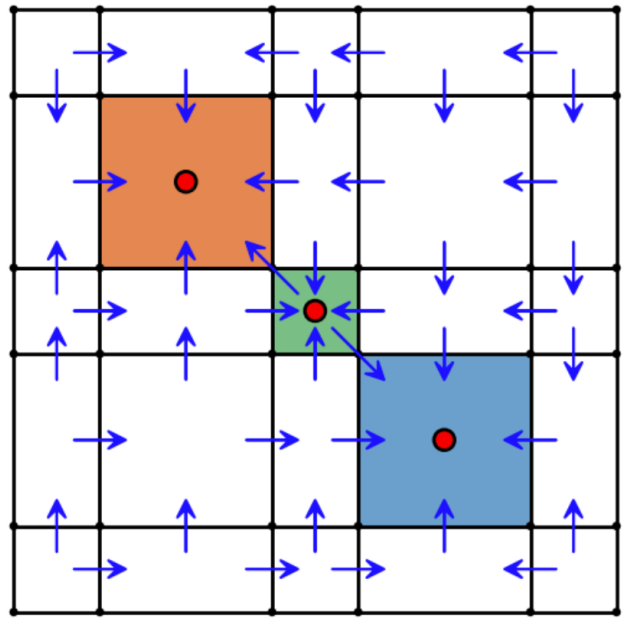
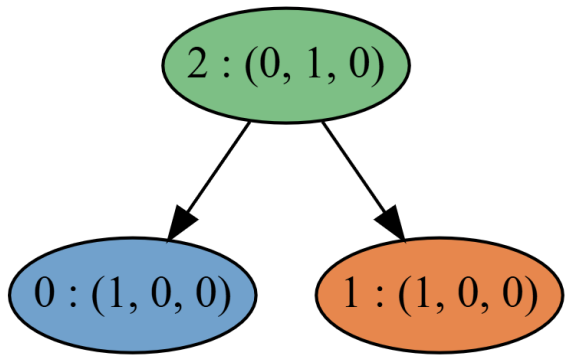
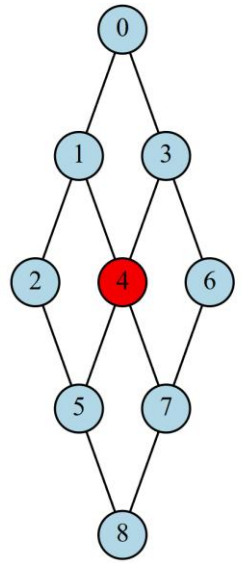
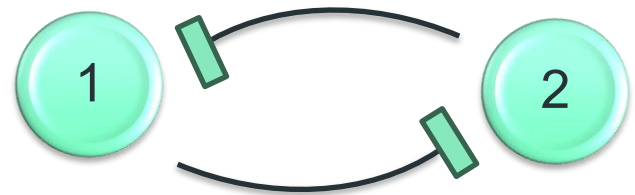
The Conley index for a source, a saddle and a sink.

<https://mat.uab.cat/~sytm/pdf/sytm-Barge.pdf>

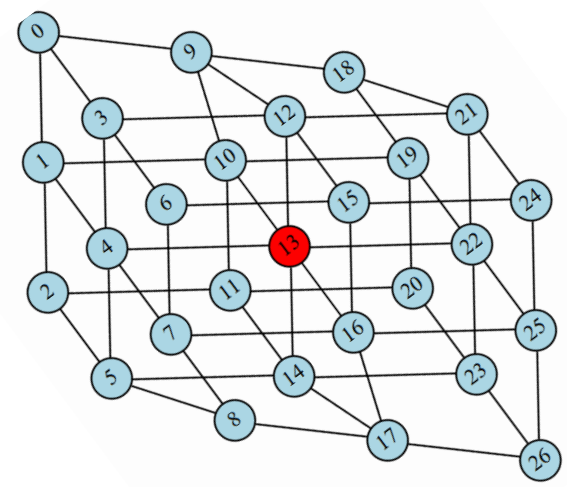
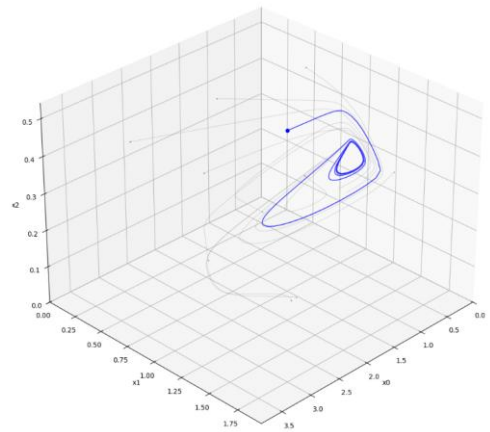
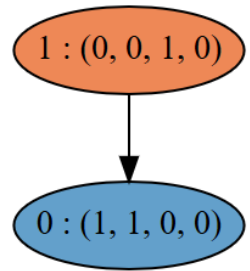
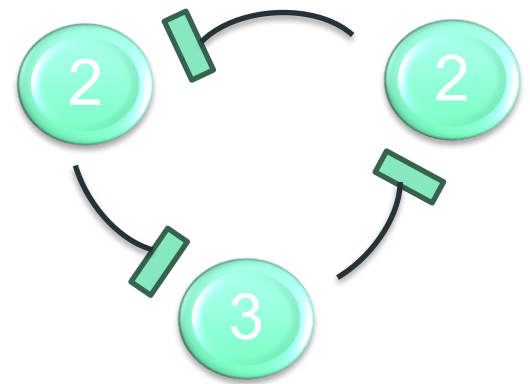
DSGRN: Dynamic signature generated by GRN.

(DSGRN) is a computational framework designed to analyze the global dynamics of gene regulatory networks without requiring precise parameter values. It partitions the high-dimensional parameter space into distinct regions and assigns each region a qualitative dynamical signature, such as a fixed point, a bistable region, or an oscillation.

Example 1: Toggle switch



Example 2



Problem Statement

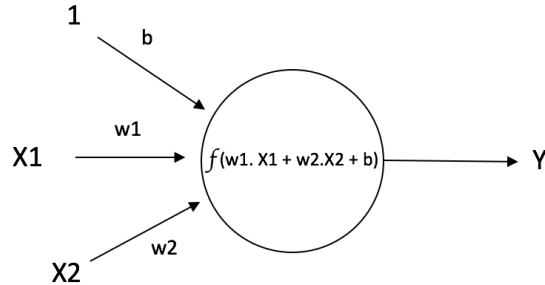
Biological data

Learn
parameters L, U, θ, d

Get Dynamics
through DSGRN



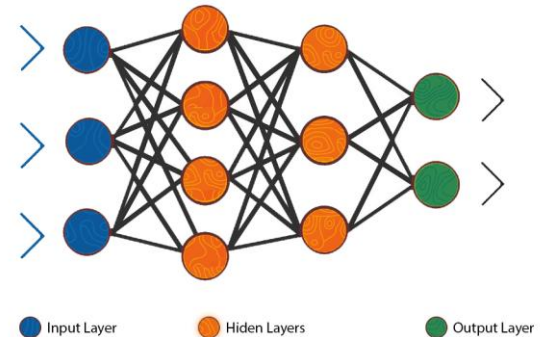
Neural Network



Output of neuron = $Y = f(w1 \cdot X1 + w2 \cdot X2 + b)$

<https://ujjwalkarn.me/2016/08/09/quick-intro-neural-networks/>

A neural network can be thought of as a parametrized function $F_{w,b}: \mathbf{R}^n \rightarrow \mathbf{R}^m$, which, under the universal approximation theorem says that, provided certain conditions are met, it can approximate any continuous function on a compact (bounded, closed) set arbitrarily well.



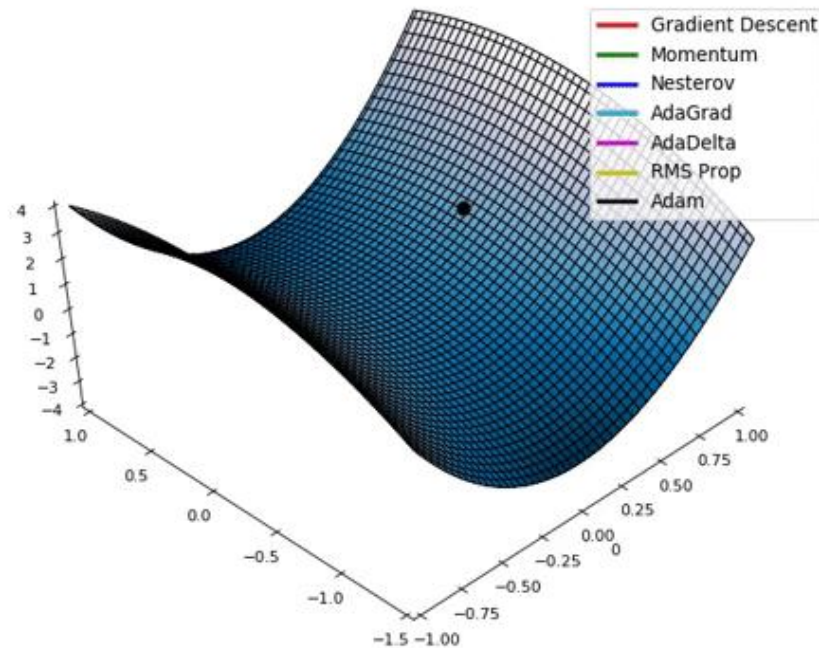
<https://businessanalytics.substack.com/p/artificial-neural-networks-anns>

Training Data:

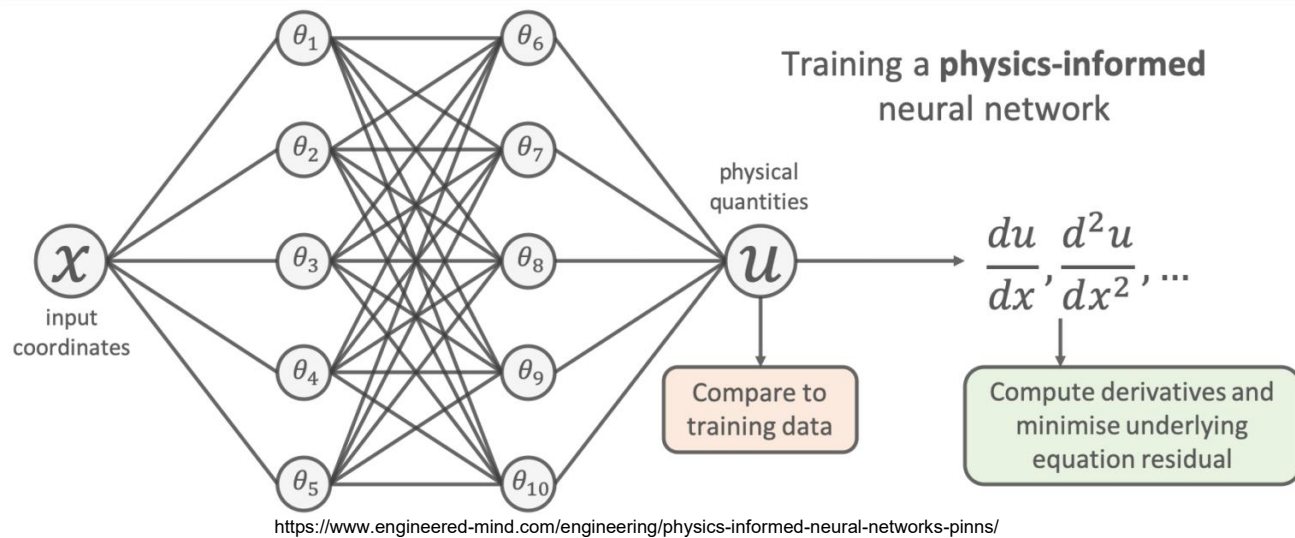
Training data: $\mathcal{D}_{train} = \{x_i, y_i\}_{i=1}^N$, where x_i is an input and y_i is its target.

Loss: given a network $F_{w,b}$, a loss function ℓ measures prediction error on one example, and training typically minimizes $\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \ell(F_{w,b}(x_i), y_i)$.

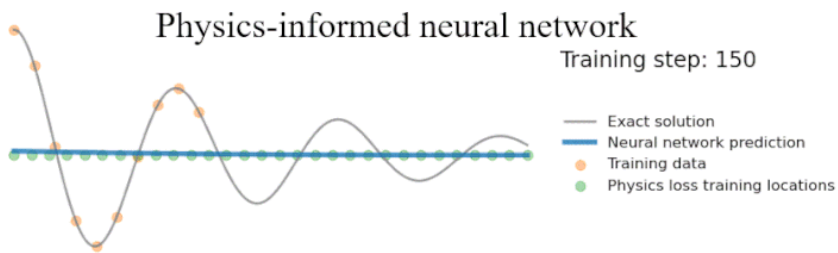
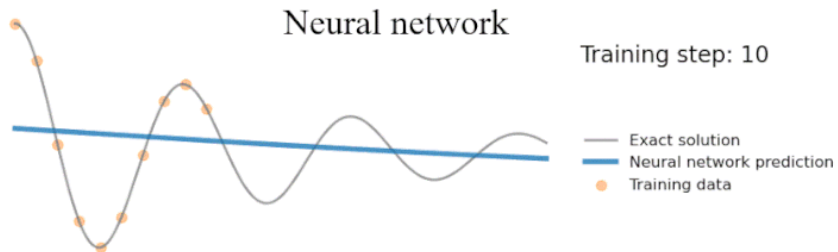
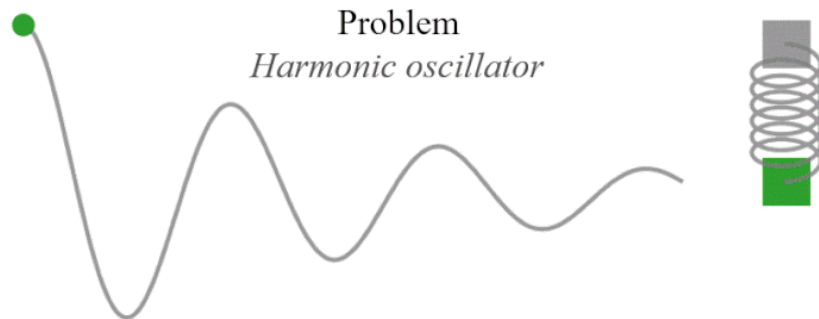
Optimizer: The goal of the optimizer is to minimize $\mathcal{L}$ by adjusting the weights w and the bias b of the neural network $F_{w,b}$



PINN: Physics informed Neural Networks



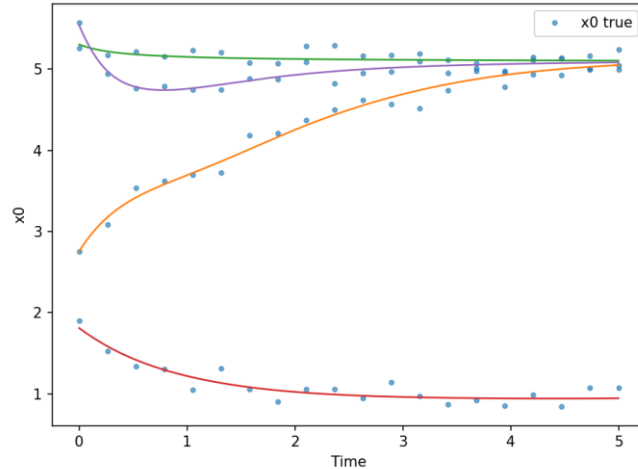
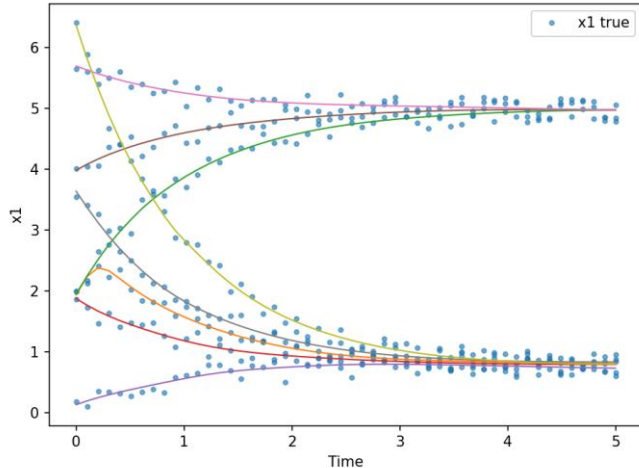
$$\mathcal{L}(w, b) = \underbrace{\frac{1}{N_d} \sum_{i=1}^{N_d} (F_{w,b}(x_i) - y_i)^2}_{\text{data term}} + \lambda_f \underbrace{\frac{1}{N_c} \sum_{j=1}^{N_c} \left(\mathcal{N}[F_{w,b}](x_j^{(c)}) \right)^2}_{\text{physics residual}} + \lambda_b \underbrace{\frac{1}{N_b} \sum_{k=1}^{N_b} \left(\mathcal{B}[F_{w,b}](x_k^{(b)}) \right)^2}_{\text{BC/IC term}}.$$



Inverse Problem:

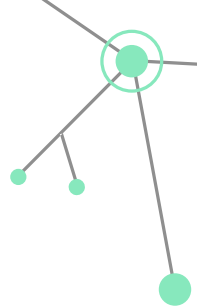
Can we estimate parameters inside our differential equations? What is the benefit of using PINN compared to least square?

YES



Spectral Bias

Standard neural networks have tendency to learn low-frequency, smooth components first. Early in training the fit can look almost linear or gently curved, and only later do higher-frequency details appear. We can solve the **spectral bias** problem by introducing feature maps. We provide hereby three types of feature maps:



Solution: Blow Up the input space

Fourier Feature map: $A \in \mathbb{R}^{b \times n}$ is a frequency matrix

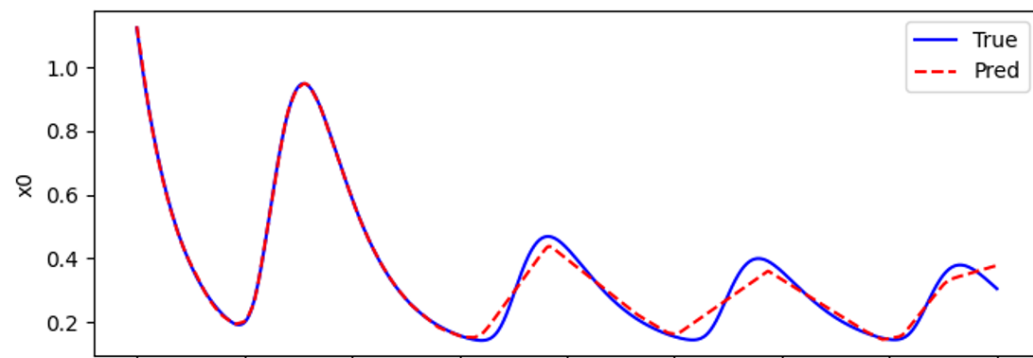
$$\gamma^1(z) = \begin{bmatrix} \cos(Az), \sin(Az) \end{bmatrix}$$

Radial Basis Function: c and σ are the centers and standard deviation respectively

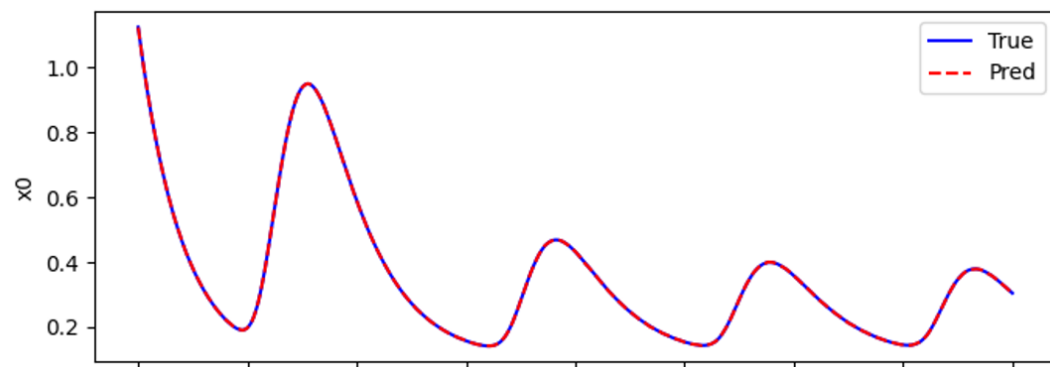
$$\gamma^2(z; c, \sigma) = \exp\left(-\frac{\|z - c\|^2}{2\sigma^2}\right)$$

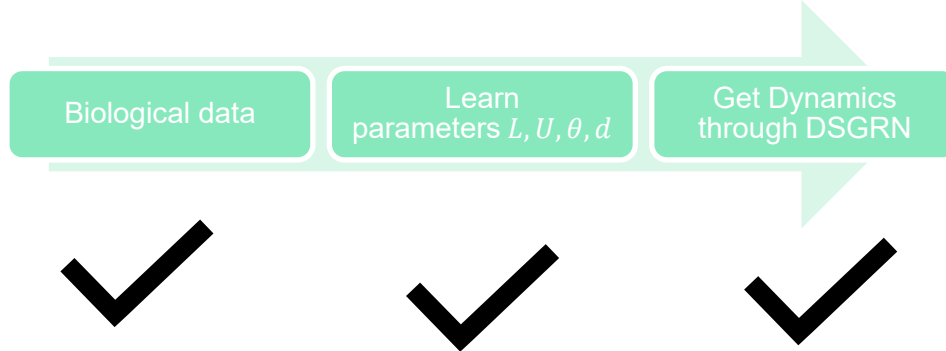
$$\left[\gamma^2(z; c_0, \sigma), \gamma^2(z; c_1, \sigma), \gamma^2(z; c_2, \sigma), \dots, \gamma^2(z; c_n, \sigma) \right]$$

No feature map used



Feature map applied



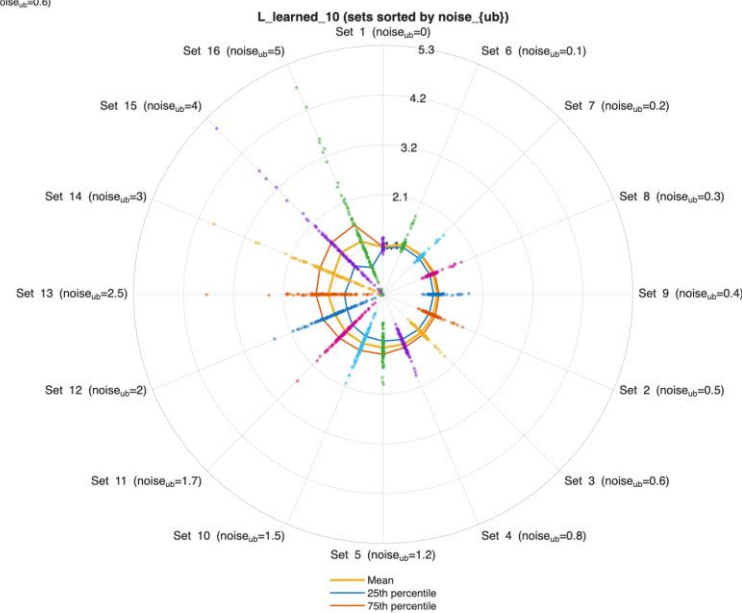
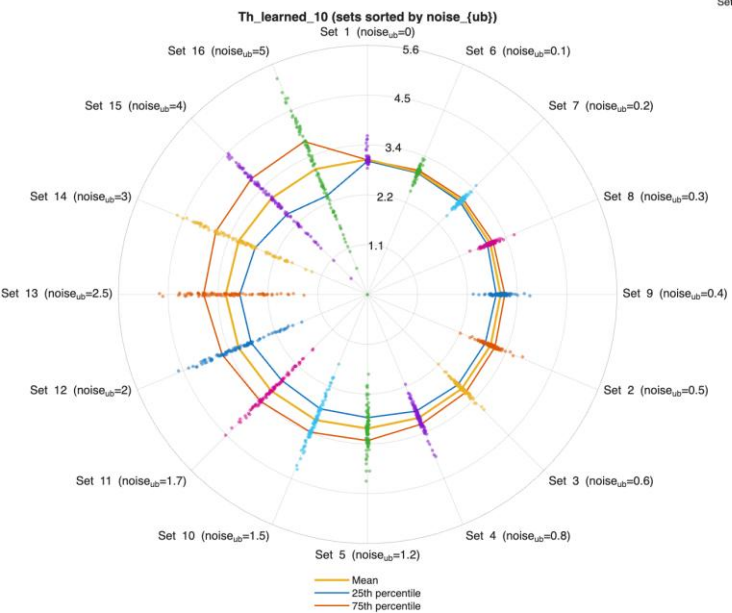
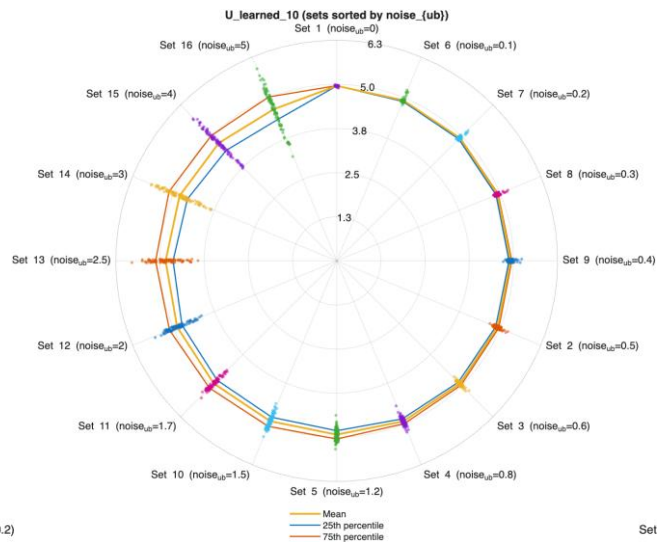


Example 1:

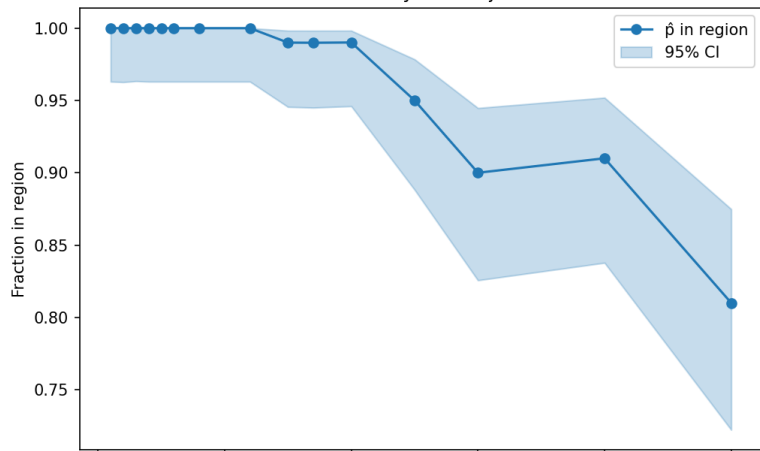
$$\frac{dx}{dt} = -x + H(y, U_{10}, L_{10}, \theta_{10}, d)$$

$$\frac{dy}{dt} = -y + H(y, U_{01}, L_{01}, \theta_{01}, d)$$

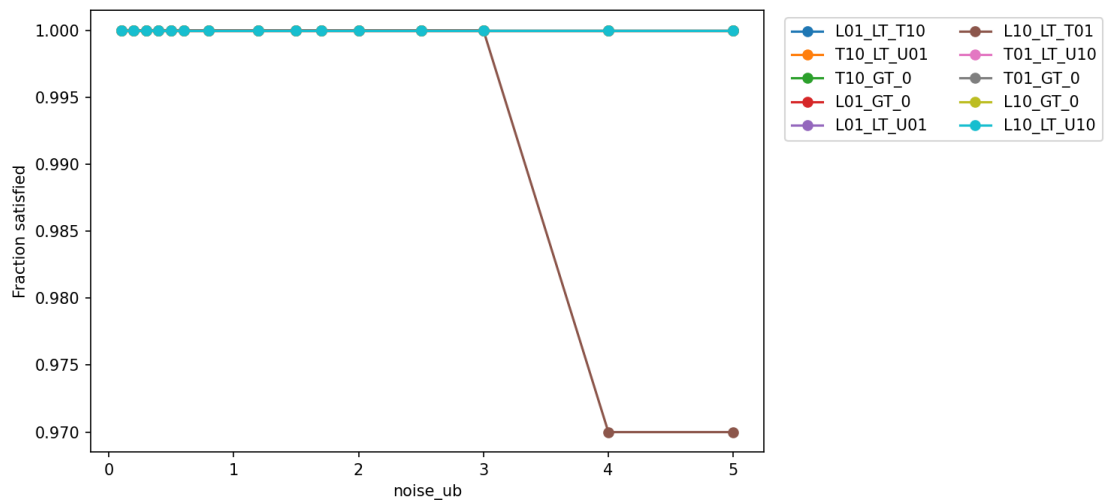
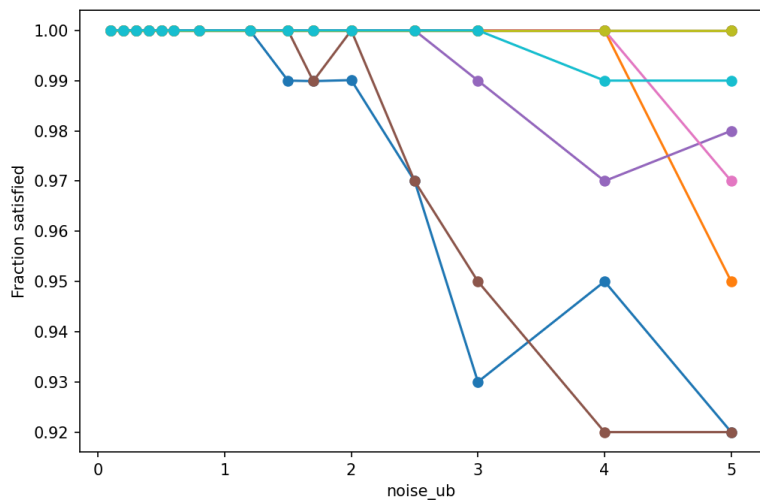
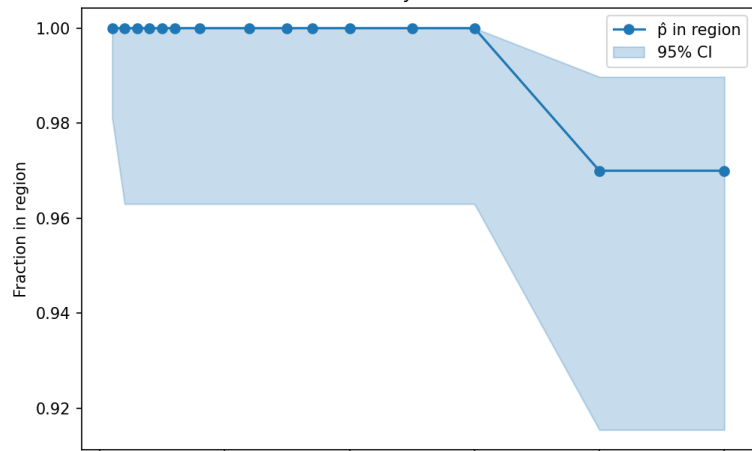
- 1) Set U_{01} and U_{10} to 5, L_{01} and L_{10} to 1, and θ_{01} and θ_{10} to 3
- 2) Simulate different set of runs based on different trajectory initial conditions, neural network initial condition, and different observational noise levels.
- 3) Can PINN learn back the right parameters? (or at least the correct combination of parameters that can give us the right dynamics?)



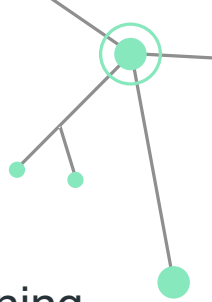
Summary: diff Traj IC



Summary: diff NN IC

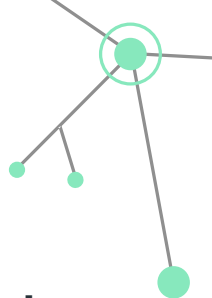


Implication of this work on Ecological and Epidemiological models?



- PINNs**: estimate model parameters from data while enforcing the governing equations
 - Morse graphs & Conley index**: give a global, qualitative map of dynamics and regime changes. **Trade-off**: computationally intensive, but offers deeper system-level understanding. Works best with difference equations/combinatorial systems.
- 

Tools to take to your SIR and PP models



Many new machine learning tools are now available and easy to implement:

Hybrid physics + learning (fit, forecast, infer)

Inverse PINNs, Neural ODEs/UDEs, Operator Learning (DeepONet, FNO)

Bayesian inference (MCMC/SMC/VI) + Data assimilation (EnKF, 4D-Var)

Discover structure + design policies (interpret, control)

SINDy + Symbolic regression (PySR, gplearn, AI Feynman, Eureqa)

Reinforcement learning / optimal control, Graph neural networks





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Thanks!

Do you have any questions?

Wael.elkhateeb@rockets.utoledo.edu



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